P.2 Solving Equations

Name _____ Period ____ Date _____

Pre-Calculus P.2 · Student sheet

I can…

- solve linear and rational equations and check for extraneous solutions.
- solve quadratic equations by factoring or extracting square roots.
- solve quadratic equations by completing the square or the Quadratic Formula.

Vocabulary · fill in the missing word

Identity — an _____ that is true for every real number in its domain

Extraneous solution — a solution of a _____ equation that does not satisfy the original equation

Quadratic equation — $ax^2 + bx + c =$ _____, $a \neq 0$

Part 1 · Four ways, one equation

Solve $x^2 - 4x - 5 = 0$ three ways.

Method	Work	Solutions
Factoring		
Completing the square		
Quadratic Formula		

Part 2 · Notes and examples (fill in each note, then work the example)

Linear equations: isolate the variable. If the variable drops out, the equation is an _____ (all reals) or has no solution.

Example 1

Solve.

a. $6(x - 1) + 4 = 3(7x + 1)$

b. $2(x + 3) = 2x + 6$

Rational equations: multiply by the LCD, then check for _____ solutions.

Example 2

Solve $\frac{1}{x - 2} = \frac{3}{x + 2} - \frac{6x}{x^2 - 4}$.

Example 3

Solve.

a. $2x^2 + 9x + 7 = 3$

b. $(x - 3)^2 = 7$

Quadratic Formula: x = _____

Example 4

Solve $x^2 + 3x = 9$. Give exact and rounded answers.

Example 5

Solve $x^2 + 6x - 5 = 0$ by completing the square.

Part 3 · Partner practice

Partner 1

Solve $3x^2 - 2x - 8 = 0$.

Partner 2

Solve $2x^2 - 4x - 3 = 0$. Round to two decimal places.